Allyson DeMerlis

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EDUCATION

2019 – 2024	**Doctor of Philosophy in Marine Biology and Ecology** University of Miami (UM) Rosenstiel School for Marine, Atmospheric, and Earth Science (Miami, FL) Advisors: Dr. Ian Enochs (NOAA) and Dr. Nikki Traylor-Knowles (UM) Dissertation: "Molecular mechanisms of enhancing coral resilience to thermal stress"
2013 – 2017	**Bachelor of Science in Biology** McGill University (Montreal, Quebec, Canada) Research: Molecular biology in *Caenorhabditis elegans*; Tropical ecology in Panama

RESEARCH EXPERIENCE

Aug 2026 – current	**Assistant Scientist**, University of Miami Rosenstiel School for Marine, Atmospheric, and Earth Science (Miami, FL) Advisor: Dr. Diego Lirman Funding: NOAA Transformational Habitat Restoration, Florida Department of Environmental Protection Project 1 (Scientist; NOAA): "Next-Generation Coral Restoration: Implementing and scaling new approaches to increase the climate resilience of Florida's Coral Reef" Project 2 (PI; DEP): "Integrating Corallivorous Fish Predation Data to Improve Coral Restoration Outcomes through Predictive Tools and Experimental Validation" Project 3 (Collaborator; NSF): "Large-scale CoPE: Reducing Climate Risks with Equitable Nature-based Solutions; Engaging Communities on Reef-Lined Coasts"
Jan 2025 – Aug 2026	**Postdoctoral Research Associate**, University of Miami Rosenstiel School for Marine, Atmospheric, and Earth Science (Miami, FL) Advisor: Dr. Diego Lirman Funding: NOAA Transformational Habitat Restoration, Florida Department of Environmental Protection Project 1 (Postdoc; NOAA): "Next-Generation Coral Restoration: Implementing and scaling new approaches to increase the climate resilience of Florida's Coral Reef" Project 2 (co-PI; DEP): "A Meta-Analysis Integrating Field Data to Quantify Corallivorous Fish Impacts on Coral Outplant Success through a Spatiotemporal Predation Pressure Index"

Aug – Oct 2024	**Postdoctoral Research Associate**, University of Miami's Cooperative Institute for Marine and Atmospheric Studies | National Oceanic and Atmospheric Administration Atlantic Oceanographic and Meteorological Laboratory (Miami, FL) Advisors: Dr. Ian Enochs and Dr. Michael Studivan Funding: NOAA Coral Reef Conservation Program and 'Omics Initiative
Aug 2019 – Jul 2024	**Graduate Research Assistant,** University of Miami Rosenstiel School for Marine, Atmospheric, and Earth Science | National Oceanic and Atmospheric Administration Atlantic Oceanographic and Meteorological Laboratory (Miami, FL) Advisors: Dr. Ian Enochs and Dr. Nikki Traylor-Knowles Funding: NOAA Coral Reef Conservation Program and 'Omics Initiative Project: "Molecular mechanisms of enhancing coral resilience to thermal stress"
May – Aug 2019	**Research Intern,** Mote Marine Laboratory's International Center for Coral Reef Restoration and Research (Summerland Key, FL) Advisors: Dr. Hanna Koch and Dr. Abigail Clark Project: "Optimizing protocol for use of 3D scanner to measure coral growth"
Feb – May 2019	**Coral Health and Disease Intern**, Mote Marine Laboratory's International Center for Coral Reef Restoration and Research (Summerland Key, FL) Advisors: Dr. Hanna Koch and Dr. Abigail Clark Project: "Development of *in-situ* spawning nursery for *Acropora cervicornis*"
Jan – Apr 2017	**Research Intern**, Parque Natural Metropolitano (Panama City, Panama) Advisor: Dr. Javier Mateo-Vega Project: "Population assessment and management strategies following introduction of invasive species of freshwater turtle, *Trachemys scripta*"
Jan – Jun 2016	**Research Intern,** Roy Lab at McGill University Department of Biological Sciences (Montreal, Canada) Advisor: Dr. Richard Roy Project: "Fatty acids as signaling molecules in lipolysis: Investigating change in lipid abundance in relation to KIN-2 and ATGL-1 activity in *C. elegans*"

PEER-REVIEWED AND PREPARED PUBLICATIONS

h-index=4; i10-index=3; Citations=172

9. **DeMerlis A,** Rodriguez K, Traylor-Knowles N. (2026) The coral innate immune system is activated in the early hours following injury. *PLOS One,* in review.
8. **DeMerlis A,** McAnally A, Bourque A, D'Alessandro M, Gilliam DS, Muller EM, Parsons KT, Sharp WC, Voss JD, Lirman D. (2026) Coral predation across multiple gradients: Implications for survivorship and reef restoration. *Scientific Reports*, in review.
7. **DeMerlis A,** Weisman E, D'Alessandro M, Tobias B, Unsworth J, Lirman D. (2026) The secret life of coral predators: Documenting fish corallivory on restored corals in Florida using Coral-Baited Remote Underwater Video Systems (C-BRUVS). *Bulletin of Marine Science.* In review.
6. **DeMerlis A,** Foord C, Hesley DJ, Karp RF, Isma LM, Andrade-Rodriguez N, Wong KH, Studivan MS, Jankulak M, Baker AC, Lirman D, Traylor-Knowles N, Enochs IC. (2026) Molecular mechanisms of *Acropora cervicornis* following transplantation to a marginal, nearshore environment. *Frontiers in Marine Science*. doi:10.3389/fmars.2026.1796814
5. **DeMerlis A,** Studivan M, Wong K, Soderberg N, Ehrens D, Isma L, Coscon K, Rosing K, Thomas R, D'Alessandro M, Dvorkin D, Unsworth J, Kiel P, Palacio-Castro A, Lirman D, Baker A, Muller E,

Traylor-Knowles N, Enochs IC (2025) Differences in molecular responses to a thermally variable preconditioning treatment for two Caribbean coral species. *Ecology and Evolution*. DOI: doi:10.1002/ece3.72108

4. Webb AE, Palacio-Castro AM, Cooke K, Eaton KR, Chomitz B, Soderberg N, Chakraborty M, Zagon Z, Boyd A, Kiel PM, **DeMerlis A**, Perry CT, Enochs IC (2024) Rubble persistence under Ocean Acidification threatened by accelerated bioerosion and lower-density coral skeletons. *Global Change Biology*. doi:10.1111/gcb.17371
3. **DeMerlis A**, Kirkland A, Kaufman ML, Mayfield AB, Formel N, Kolodziej G, Manzello DP, Lirman D, Traylor-Knowles N, Enochs IC (2022) Pre-exposure to a variable temperature treatment improves the response of *Acropora cervicornis* to acute thermal stress. *Coral Reefs*. doi:10.1007/s00338-022-02232-z
2. Traylor-Knowles N, Connelly MT, Young BD, Eaton K, Muller EM, Paul VJ, Ushijima B, **DeMerlis A,** Drown MK, Goncalves A, Kron N, Snyder GA, Martin C, Rodriguez K (2021) Gene expression response to stony coral tissue loss disease transmission in *M. cavernosa* and *O. faveolata* from Florida. *Frontiers in Marine Science*. doi:10.3389/fmars.2021.681563
1. Koch HR, Wallace B, **DeMerlis A**, Clark AS, Nowicki RJ (2021) 3D scanning as a tool to measure growth rates of live coral microfragments used for coral reef restoration. *Frontiers in Marine Science.* doi:10.3389/fmars.2021.623645

GRANTS AND AWARDS

2026	Florida Department of Environmental Protection Coral Protection and Restoration Program (PI; $234,000)
2026	University of Miami Rosenstiel School Research Day "Best Visual Design Award" ($250)
2025	Georgia Tech Seed Grant (Collaborator; $5,000)
2025	Florida Department of Environmental Protection Coral Protection and Restoration Program (PI; $147,000)
2024	NOAA Oceanic and Atmospheric Research Team Member of the Year
2024	Evolving Seas RCN MarineOmics Hack-A-Thon ($500)
2022	University of Miami Graduate Student Organization Travel Fund ($500)
2022	University of Miami Graduate Student Association Academic Excellence Award ($650)
2022	Capt. Harry D. Vernon, Jr. Scholarship ($200)
2020	David Rowland Fellowship ($3500)
2019	Research Experience for Undergraduates Scholarship ($6000)
2017	McGill University Dean's Multidisciplinary Undergraduate Research List

TEACHING ASSISTANTSHIPS

2021	Saltwater Semester: Cnidarian Microbiology and Immunology, University of Miami
2020	Marine Biology Laboratory, University of Miami

ORAL PRESENTATIONS

2026	**DeMerlis A,** Unsworth J, Tobias B, Lachnit C, D'Alessandro M, Hightshoe M, Niedermaier Z, Gilliam D, Walker B, Lirman D. "Integrating Coral Predation Risk into Site Selection for Restoration of Florida's Coral Reef." Reef Futures, Playa del Carmen, Mexico
2026	**DeMerlis A,** Unsworth J, Tobias B, Lachnit C, Paulsen-Sacks S, Bilder S, D'Alessandro M, Hightshoe M, Niedermaier Z, Gilliam D, Walker B, Lirman D. "Predicting Coral

	Predation Along Florida's Coral Reef: A Decadal Synthesis, Video Monitoring, and a Spatial Predation Pressure Index." ReeFlorida, Miami, Florida, USA
2026	**DeMerlis A,** Unsworth J, Tobias B, Lachnit C, Weisman E, D'Alessandro M, Lirman D. "Toward predicting corallivorous fish predation in coral restoration on Florida's Coral Reef." International Coral Reef Symposium, Auckland, New Zealand
2026	**DeMerlis A,** McAnally A, Unsworth J, Tobias B, Lachnit C, Weisman E, D'Alessandro M, Lirman D. "Towards predicting coral predation along Florida's Coral Reef to improve efficacy of massive coral outplanting." University of Miami Rosenstiel Research Day, Miami, Florida, USA
2025	**DeMerlis A,** McAnally A, Unsworth J, Tobias B, Tenberg D, Fitzgerald M, Weisman E, D'Alessandro M, Lirman D. "Predicting coral predation along the Florida Reef Tract: A synthesis of decadal research and restoration efforts." ReeFlorida, Miami, Florida, USA
2024 (accepted)	**DeMerlis A,** Studivan M, Wong K, Snyder G, Isma L, D'Alessandro M, Palacio-Castro A, Lirman D, Traylor-Knowles N, Enochs IC. "Evaluating an ex-situ variable temperature treatment as a stress-hardening technique for reef-building corals." Reef Futures, Riviera Maya, Mexico
2024 (accepted)	**DeMerlis A,** Studivan M, Besemer N, Jankulak M, Kolodziej G, Boyd A, Soderberg N, Dennison C, Karp R, Baker A, Traylor-Knowles N, Enochs IC. "Life on the edge: Investigating coral resilience in marginal environments." European Coral Reef Symposium, Naples, Italy
2023	**DeMerlis A**, Studivan M, Snyder G, Soderberg N, Ehrens D, Isma L, Rosing K, Coscon K, Thomas R, Dvorkin D, Chakraborty M, Kiel P, Babbitz H, Unsworth J, D'Alessandro M, Palacio-Castro A, Lirman D, Traylor-Knowles N, Enochs IC. "Evaluating the efficacy of a variable temperature treatment to stress-harden threatened reef-building corals in Florida." Benthic Ecology Meeting, Miami, Florida, USA
2022	**DeMerlis A**, Kirkland A, Kaufman ML, Mayfield AB, Formel N, Kolodziej G, Manzello DP, Lirman D, Traylor-Knowles N, Enochs IC. "Variable temperature treatments alter a coral's response to acute thermal stress." International Coral Reef Symposium, Bremen, Germany
2022	**DeMerlis A**, Kirkland A, Kaufman ML, Mayfield AB, Formel N, Kolodziej G, Manzello DP, Lirman D, Traylor-Knowles N, Enochs IC. "Prolonged exposure to a variable, elevated temperature regime as a tool for stress-hardening nursery-raised *Acropora cervicornis*." Reef Futures, Key Largo, Florida, USA
2022	**DeMerlis A,** Beasley S, Rodriguez K, Traylor-Knowles N. "Early-hour molecular signatures of wound healing in *Pocillopora damicornis*." Cnidofest, Davis, California, USA
2021	**DeMerlis A**, Kirkland A, Kaufman ML, Mayfield AB, Formel N, Kolodziej G, Manzello DP, Lirman D, Traylor-Knowles N, Enochs IC. "Pre-exposure to variable temperature treatments improves the response of *Acropora cervicornis* to acute thermal stress." International Coral Reef Symposium (virtual)

INVITED LECTURES

2021–2024	Marine Biology and Ecology Department Seminar at UM Rosenstiel School
2022	Biology Course at Yale University
2020	Boston University Marine Program
2019	Osher Lifelong Institute, University of Miami
2019	Mote Marine Laboratory's International Center for Coral Reef Research and Restoration

MENTORING

2026	Stella Bilder, University of Miami Master of Science
2026	Victoria Lahey, University of Miami Undergraduate
2025	Megan Pike, University of Miami Master of Professional Science
2025	Sydney Rodetsky, University of Miami Master of Professional Science
2025	Rowan Anderson, University of Miami Master of Professional Science
2025	Mason Fitzgerald, University of Miami Master of Science
2024	Erin Weisman, University of Miami Master of Science
2023	Remedy Rule, University of Miami Master of Science
2022	Katrina Cocson, University of Miami Undergraduate
2022	Katrina Rosing, University of Miami Master of Professional Science
2022	Samantha Beasley, University of Miami Master of Professional Science

PROFESSIONAL SERVICE AND ASSOCIATIONS

Reviewer:	*Journal of Experimental Biology, Coral Reefs, Scientific Reports, PeerJ, BMC Biology*, National Science Foundation Biological Oceanography Program
Associations:	International Society for Reef Studies

EDUCATIONAL OUTREACH

2021–current	"Rescue a Reef": citizen science coral restoration program
2022	University of Miami Rosenstiel School Ocean Explorers event
2022	Scientists For Every School in Florida presentations (virtual)
2021	University of Miami Rosenstiel School Graduate Program Open House (virtual)
2020	University of Miami Rosenstiel School "Students for Students" Program (virtual)
2019–2022	"Skype a Scientist" presentations (virtual)
2019	Mote Marine Laboratory Ocean Fest

CERTIFICATIONS

2025	DAN CFA Pro First Aid
2020	Motorboat Operators Certification Course (MOCC)
2020	NOAA Small Boat Component Course
2019	AAUS (American Academy of Underwater Sciences) Diver (60ft; 300 dives)
2018	PADI Rescue Diver
2018	PADI Enriched Air Diver
2018	PADI Advanced Open Water Diver
2018	PADI Open Water Certification
2018	FWC Florida Boating Safety Education Identification Card

WORKSHOPS

2026	SEFCRI Team Meeting, Harbor Branch, FL
2026	NOAA Resilience Restoration Annual Collaboration Workshop, Miami, FL
2026	University of Miami AI-Assisted Research Skills Workshop, Miami, FL
2025	NFWF SeaFoundry Data Management Tools for Coral Restoration, Key Largo, FL
2025	NOAA Resilience Restoration Annual Collaboration Workshop, Miami, FL

2025	DEP FCR Propagation and Restoration Technical Meeting, West Palm Beach, FL
2023	Dallas Murphy Scientific Writing Workshop, Miami, FL
2019	University of Miami Libraries Data Management Workshop, Miami, FL

PRESS AND RESEARCH FEATURES

1. **South Florida got a (small) break from record heat in June. Don't get used to it**. (2024). *Miami Herald.*
2. **Water temps soar in Florida Keys. Can coral reefs stand another bout of extreme heat?** (2024). *Miami Herald.*
3. **South Florida waters hit hot tub level and may have set world record for warmest seawater.** (2023) *CBS News Miami.*
4. **Florida's rising water temperatures raise concerns for health of coral reefs, scientists say.** (2023). *Reuters.*
5. **With 'Mission: Iconic Reefs', NOAA aims to restore Florida Keys with climate-resilient corals.** (2023). *NOAA Climate.gov.*
6. **Corals can be 'trained' to tolerate heat stress, study finds**. (2022) *University of Miami Rosenstiel School News.*
7. **New technique shows promise to restore Florida's threatened coral.** (2022) *NBC News WFLA Tampa.*

SKILLS

- Scientific Diving, Field Work, Scientific Boating and Surface Support
- Programming Languages: R, Unix Shell, Python
- Transcriptomics
- Visual Software: Adobe Illustrator, Adobe Photoshop, Canva
- Image Software: ImageJ, FlexScan3D
- Data Software: Microsoft Office Suite (Excel), RStudio
- Laboratory Techniques: DNA and RNA extractions, cDNA library preparation, qRT-PCR, Fluorescence automated cell sorting (FACS)
- Coral Husbandry